Social Media Impact on Teen Mental Health

Predicting Behavioral and Depression Risk Using Machine Learning

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Problem & Dataset

Problem

Why this matters: Teen mental health concerns are increasing, while social media usage continues to rise.

Objective: Use daily behavioral indicators such as social media usage, sleep habits, screen time, and lifestyle factors to identify patterns associated with depression risk.

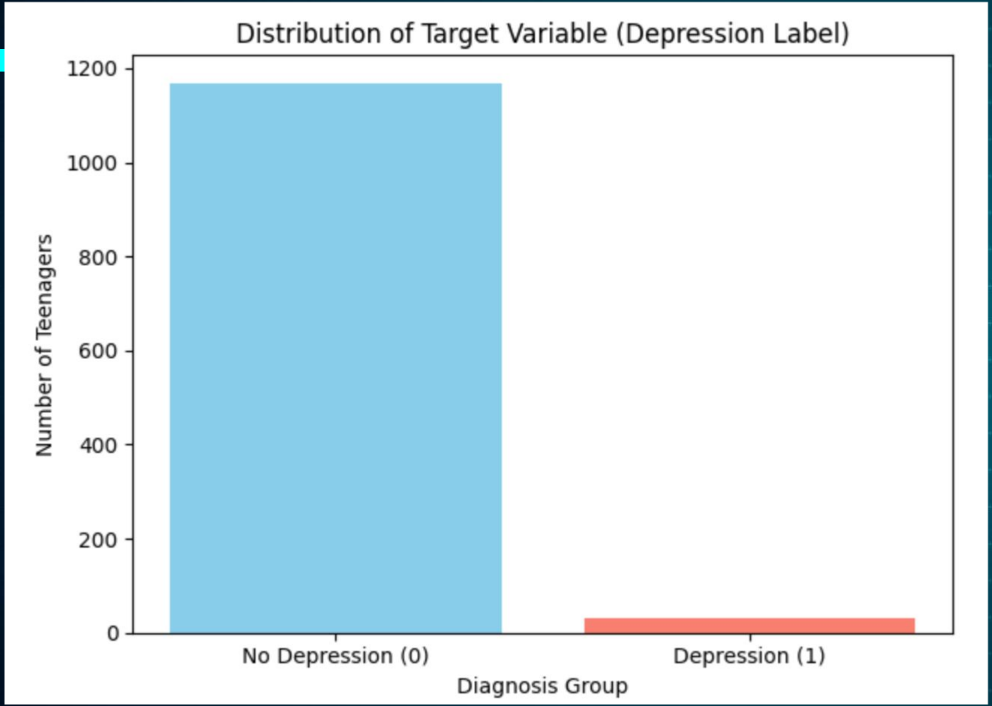
Problem Framing: To frame a supervised binary classification task that predicts whether a teenager is at risk for depression (0 = No, 1 = Yes) based on daily habits.

Dataset

age	gender	daily_social_media_hours	platform_usage	sleep_hours	screen_time_before_sleep	academic_performance	physical_activity	social_interaction_level	stress_level	anxiety_level	addiction_level	depression_label
14	male	7.9	Instagram	7.4	2.9	3.01	1.5	low	2	2	1	0
19	female	1.9	TikTok	8	2.9	3.22	0.8	high	8	1	10	0
17	female	1.3	Instagram	7.6	0.5	3.92	0	high	2	4	2	0
15	male	7.4	TikTok	6.9	1.6	3.48	0.8	medium	1	7	9	0
15	female	4.7	Both	4.9	3	2.37	1.4	medium	3	5	2	0
19	female	7.4	Both	4.4	2.4	2.63	0.6	high	3	5	7	0
18	female	2.5	Instagram	6.4	2.4	2.63	0.7	low	2	2	5	0
16	male	4	Both	4.2	0.5	2.4	1.3	low	6	10	5	0
19	female	3.3	TikTok	5	2.1	2.04	0.9	high	1	10	9	0
15	male	1.9	TikTok	4.9	1.5	3.77	1.1	high	1	1	4	0
16	male	7.2	Both	5.8	2.9	3.16	1.2	medium	5	8	9	0
13	male	4.9	Instagram	4.7	2.5	3.92	1.2	low	1	10	8	0
16	male	6.2	Both	7.2	2.6	2.02	1.7	medium	5	7	6	0
17	female	6.8	TikTok	5.7	1.1	3.29	1.3	medium	1	10	10	0
15	male	5.7	TikTok	5.9	0.8	3.5	0.7	low	1	9	1	0
14	female	3.6	Both	5.3	0.6	2.99	0.3	high	4	10	2	0
13	male	1.5	Both	4.2	1.4	2.59	0.5	high	4	5	8	0
16	male	3.1	Both	6.1	0.8	2.11	1.9	high	10	7	10	0
16	female	6.7	Both	6.8	1.9	3.08	1.4	high	10	9	1	0
14	female	1.2	Instagram	7.6	1	3.85	0.8	high	8	4	6	0
16	male	3.4	TikTok	4.6	2.1	2.24	0.4	high	4	1	3	0

- **Size:** 1,200 rows x 13 columns
- **Target Variable (y):** depression_label (Binary Classification)
- **Feature Types (X):**
  - Categorical: gender, platform_usage (Instagram, TikTok, Both), and social_interaction_level.
  - Numerical: age, daily_social_media_hours, sleep_hours, screen_time_before_sleep, academic_performance, physical_activity, stress_level, anxiety_level, and addiction_level

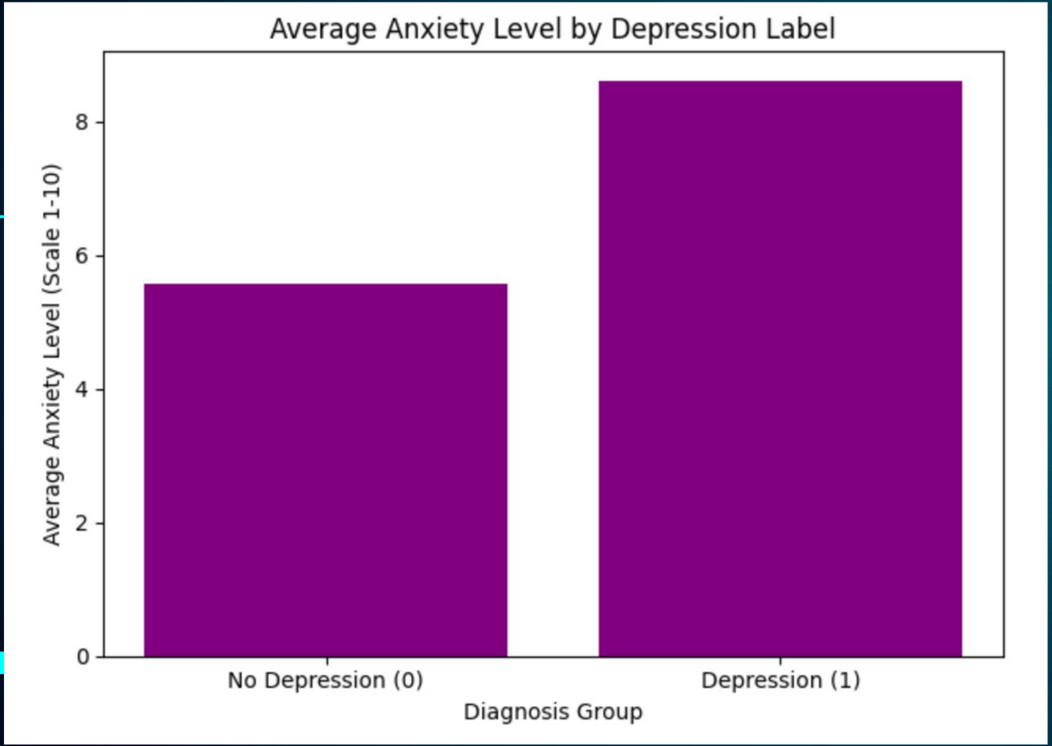
Exploratory Data Analysis (EDA)



Class Imbalance Challenge

- **No Risk (0):** 1,169 teens
- **At Risk (1):** 31 teens
- Predicting "No Risk" for everyone gives 97.4% accuracy but identifies 0 at-risk teens.
- **Recall** is more important than accuracy in this case.

Average Anxiety Level by Depression Label



- Higher anxiety levels were observed among at-risk (1) teens.
- Anxiety shows a strong relationship with depression risk.
- Anxiety may be an important predictor in the model.

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Pipeline

The Leakage-Safe Preprocessing Pipeline

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42, stratify=y)
```

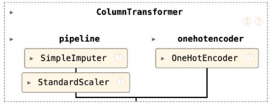
Step 1: Train/Test Split

```
numeric_cols = ['age', 'daily_social_media_hours', 'sleep_hours', 'screen_time_before_sleep', 'physical_activity', 'academic_performance', 'stress_level', 'anxiety_level', 'addict
```

```
categorical_cols = ['gender',
    'platform_usage',
    'social_interaction_level']
```

```
preprocess = make_column_transformer ((make_pipeline (SimpleImputer() , StandardScaler()), numeric_cols), (OneHotEncoder(handle_unknown = 'ignore'), categorical_cols ) , )
```

```
preprocess
```



Step 2: ColumnTransformer Configuration

- Numeric Pipeline
- Categorical Pipeline

Models Compared

K-Nearest Neighbors (KNN)

```python
# Model 1:  KNeighborsClassifier
model = make_pipeline (preprocess, KNeighborsClassifier())
```

Logistic Regression

```python
# Model 2: LogisticRegression
model_lr = make_pipeline (preprocess, LogisticRegression(class_weight='balanced', random_state=42))
```

Cross-Validation Comparison

```
knn_cv_scores = cross_val_score(model, X_train, y_train, cv=5)

knn_mean_cv = knn_cv_scores.mean()

knn_mean_cv

np.float64(0.9739583333333334)

lr_cv_scores = cross_val_score(model_lr, X_train, y_train, cv=5)

lr_mean_cv = lr_cv_scores.mean()

lr_mean_cv

np.float64(0.94375)

# Logistic Regression chosen: lower accuracy (94.38%) but handles class imbalance safely.
```

- KNN mean CV: 97.4%
- LR mean CV: 94.4%

- Chosen Model: Logistic Regression
- Reason: Better for identifying rare at-risk cases.

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Results

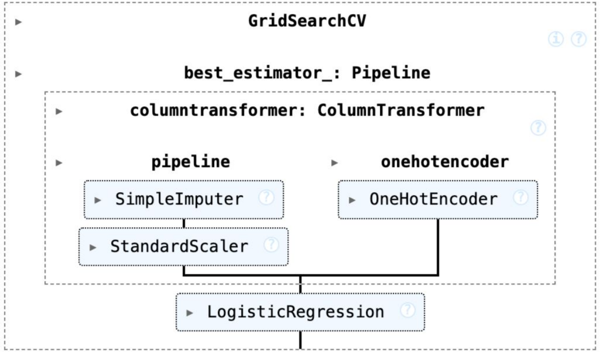
Hyperparameter Tuning

- Method: GridSearchCV (5-fold CV)
- Tested C: 0.01, 0.1, 1, 10, 100
- Best Parameter: {'logisticregression__C': 100}.
- Best CV Accuracy: **95.10%**

```
param_grid_lr = {'logisticregression__C': [0.01, 0.1, 1, 10, 100]}
```

```
grid_search_lr = GridSearchCV(estimator=model_lr, param_grid=param_grid_lr, cv=5, scoring='accuracy')
```

```
grid_search_lr.fit(X_train, y_train)
```



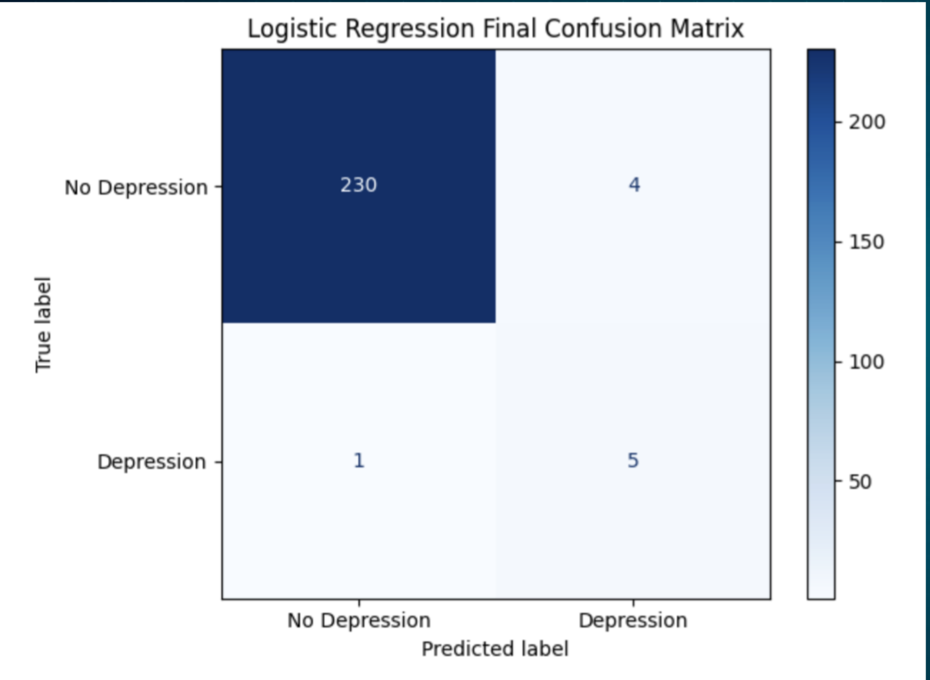

```
grid_search_lr.best_params_
```

```
{'logisticregression__C': 100}
```

```
grid_search_lr.best_score_
```

```
np.float64(0.9510416666666666)
```

Final Test Set Results



Accuracy: 97.92%

Beats the naive majority-class baseline of 97.42%

Precision: 55.56%

Correctly identified 5 out of 6 actual depression cases in the test set

Recall: 83.34%

4 out of 9 predicted cases were false positives

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Takeaways

Takeaways, Limitations, & Next Steps

Takeaways

- Anxiety & stress were top predictors.
- More social media, less sleep = higher risk.
- Logistic Regression performed best.

Limitations

- Highly imbalanced dataset (31 at-risk teens out of 1,200).
- Limited number of depression cases made prediction more challenging.

Next Steps

- Collect more data from at-risk teens to create a more balanced dataset.
- Test additional models and features to improve prediction performance.

Thank you!

Do you have any questions?

Sources

- Kaggle. Teenager Mental Health Dataset. Kaggle, https://www.kaggle.com/datasets/algozee/teenager-menthal-healy. Accessed 9 June 2026.